8. Project Demonstration

Project Title

OptiCrop – Smart Agricultural Production Optimization Engine

Demonstration Overview

The demonstration presents the complete workflow of the project: data collection and preprocessing, model training and evaluation, and the full Flask web application.

Demonstration Steps

Step 1

Launch the Flask application (python app.py) and open http://127.0.0.1:5000/ in a browser.

Step 2

Show account creation at /register (including the role dropdown and password show/hide toggle), then sign in at /login.

Step 3

On the Overview page, show the OptiCrop dashboard and navigation sidebar.

Step 4

Go to Smart Recommendation and enter Nitrogen (N), Phosphorous (P), Potassium (K), Temperature (°C), Humidity (%), Soil pH and Rainfall (mm), then click "Recommend crop." Show the recommended crop, its confidence percentage, and the generated report, then click "Download PDF report" and open the downloaded file to show the full styled report.

Step 5

Demonstrate Suitability Assessment (select a crop and submit conditions) and Crop Library (browse crop cards with season, temperature, pH and water requirement).

Step 6

Show the Research & Policy dashboard (prediction-volume trend and top-predicted-crops chart).

Step 7

Go to Prediction History, search by crop, and click "Export CSV" to show the download matches exactly the searched/filtered rows.

Screenshots to Include

- Register page (role dropdown, password toggle)
- Login page (password toggle, Forgot password link)
- Overview / Dashboard
- Smart Recommendation form and result, with Download PDF report button
- Opened PDF report

- Suitability Assessment form and report
- Crop Library
- Research & Policy dashboard
- VS Code project structure
- Terminal showing Flask running
- GitHub repository

Future Enhancements

- Live Weather API integration.
- Fertilizer recommendation system.
- Deploy the application on a cloud platform.
- Interactive dashboards and richer analytics.

Conclusion

The demonstration successfully demonstrates the application of Machine Learning and Artificial Intelligence in agriculture. The system analyzes soil nutrients and environmental conditions to recommend the most suitable crop, enabling users to make informed and data-driven farming decisions. With its user-friendly interface, crop suitability assessment, research dashboard, crop library, and prediction history, the application provides a comprehensive solution for smart agriculture. Overall, the project promotes sustainable farming practices, improves resource utilization, and contributes to enhanced agricultural productivity through intelligent crop recommendation.